

YIHENG SHEN

CONTACT INFORMATION

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EDUCATION

Duke University *August 2021 - May 2026 (Expected)*
Ph.D. Candidate, Computer Science
Advisor: Professor Kamesh Munagala

Tsinghua University *August 2017 - July 2021*
B.Eng., Institute for Interdisciplinary Information Sciences (Yao Class)

RESEARCH INTEREST

My research interests lie in the interface of economics and computation, especially fairness and stability in resource allocation problems.

RESEARCH EXPERIENCE

Harvard University *February 2020 - August 2020*
Visiting scholar in Professor Yiling Chen's group. We mainly worked on Information Pricing and Data Purchasing.

The University of Hong Kong *July 2019 - August 2019*
Research intern in Professor Zhiyi Huang's group. We worked on sample complexity in auction design.

Tsinghua University *2018 - 2020*
Advisor: Professor Pingzhong Tang, IIIS. We mainly worked on mechanism design for multi-party machine learning.

INDUSTRY EXPERIENCE

Meta, Menlo Park *May 2025 - Aug 2025*
I was a software engineer intern in the Facebook Feed Ranking team, hosted by Wenshun Liu. I mainly worked on feature analysis for the core in-feed recommendation ranking model.

Google Research, Mountain View *June 2023 - November 2023*
I was a student researcher in the Geo-algorithms team, hosted by Aaron Schild. We mainly worked on algorithms in traffic routing. I extended my internship as a part-time student researcher till November 2023.

CONFERENCE PUBLICATIONS

The Price of Competitive Information Disclosure.
With *Siddhartha Banerjee*, *Kamesh Munagala*, and *Kangning Wang*. **STOC 2026**

The Limits of Interval-Regulated Price Discrimination.
With *Kamesh Munagala* and *Renzhe Xu*. **WINE 2025 (Best Paper Award)**

Fair Division via the Cake-Cutting Share.

With *Yannan Bai, Kamesh Munagala* and *Ian Zhang*. **AAAI 2025 (Oral Presentation)**

Majorized Bayesian Persuasion and Fair Selection.

With *Siddhartha Banerjee, Kamesh Munagala*, and *Kangning Wang*. **SODA 2025**

First Passage Percolation with Queried Hints.

With *Kritikorn Karntikoon, Sreenivas Gollapudi, Kostas Kollias, Aaron Schild* and *Ali Sinop*. **AIS-TATS 2024**

Fair Price Discrimination.

With *Siddhartha Banerjee, Kamesh Munagala* and *Kangning Wang*. **SODA 2024**

Fair Multiwinner Elections with Allocation Constraints.

With *Ivan-Aleksandar Mavrov* and *Kamesh Munagala*. **EC 2023**

Auditing for Core Stability in Participatory Budgeting.

With *Kamesh Munagala* and *Kangning Wang*. **WINE 2022**

Approximate Core for Committee Selection via Multilinear Extension and Market Clearing.

With *Kamesh Munagala, Kangning Wang* and *Zhiyi Wang*. **SODA 2022**

Targeting Makes Sample Efficiency in Auction Design.

With *Zhiyi Huang, Xiangning Wang* and *Yihang Hu*. **EC 2021**

Truthful Data Acquisition via Peer Prediction.

With *Shuran Zheng* and *Yiling Chen*. **NeurIPS 2020**

JOURNAL PUBLICATION

A mechanism design approach for multi-party machine learning.

With *Mengjing Chen, Yang Liu, Weiran Shen, Pingzhong Tang* and *Qiang Yang*. **Theoretical Computer Science (TCS)**

TEACHING

Teaching Assistant: COMPSCI 230, Discrete Mathematics

Spring 2022, Spring 2023

Teaching Assistant: COMPSCI 535, Algorithmic Game Theory

Spring 2024

AWARD

Best Preliminary Exam Award, Duke Computer Science

November 2024